



CSC1109 – Assignment 2: Mid-way report

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Dataset

Using data from public TfL (Transport for London) data to do with Santander bikes. These are rental bikes and are collected and left at different hubs across London. We plan to merge 36 datasets to give us one large dataset; we will be looking at journeys made in 2023, so dataset '350JourneyDataExtract...' to '386JourneyDataExtract...'. Each dataset has the following columns:

- **Number:** A unique identifier for each trip
- **Start date:** date and time when trip began in format dd/mm/yyyy hh:mm
- **Start station number:** identifier for the starting station
- **Start station:** name of the starting station
- **End date:** date and time when trip ended in format dd/mm/yyyy hh:mm
- **End station number:** identifier for the ending station
- **End station:** name of the ending station
- **Bike number:** unique identifier for the bicycle used
- **Bike model:** model of the bicycle used
- **Total duration:** total time duration of the trip in the format of minutes and seconds
- **Total duration (ms):** total time duration of the trip in milliseconds

Site where we found the data: <https://cycling.data.tfl.gov.uk/>

Example dataset: [350JourneyDataExtract26Dec2022-01Jan2023.csv](#)

Technology

For this project, we plan to use Apache Spark as our main Big Data processing technology. Our dataset is approximately 1GB, so Spark is suitable, as it is designed to handle large-scale batch processing.

Spark supports efficient, distributed processing for the tasks we plan to carry out, such as merging a large number of files, filtering high numbers of rows and performing complex aggregations.

SQL querying in Spark is fast in executing queries on structured data, allowing us to run analytical queries efficiently. The fault tolerance and lazy evaluation properties of Spark also helped us choose this technology, as it will be reliable and perform well. Any failures will be automatically recovered, and our transformations will only be computed when necessary.

We plan to use Spark MLlib for machine learning, enabling us to conduct our prediction tasks, such as demand forecasting, and potentially apply methods like K-means clustering. Spark provides a single engine that supports multiple programming languages, making it a flexible choice for our project.

Analysis Plan

We want to identify peak demand times across different stations to allow for planning of bike distribution; this includes identifying the busiest times of day, days of the week and seasonal changes. We also want to explore popular routes to help inform bike lane or other infrastructure planning.

Another area to look into is the popularity and use of different stations; this could include the likelihood of bikes often being left at a station but not taken.

In terms of predictions, we want to forecast demand by predicting high-demand periods in different stations along with the likelihood that a customer will return a bike to the same station.

Team Roles & Tasks

Emma	☐ Cleaning, fixing types, timestamps, missing values ☐ Write Spark transformations, SQL queries & feature extractions ☐ Station usage analysis/queries ☐ Design & train prediction model
Maria	☐ Data ingestion (all 36 datasets) & prepare data for usage in Spark ☐ Peak demand analysis/queries ☐ Visualise results of outputs ☐ Evaluate and improve prediction model
Shared tasks	☐ Attend project meeting ☐ Write-up of methodology & results ☐ Peer review of each other's code ☐ Interpretation of findings ☐ Final report ☐ Final presentation/demo